

Ontology of Becoming

the argument

Primitive. A *law* is a map m on a carrier. Write $\text{Fix}m$ for “ m , read as an equation, has a solution” ($\exists x. x = m x$), and $\text{Run}m$ for “ m , read as a recursion, is solved from every seed, and every term of every solution differs from its successor.”

Df. m is *structureless* when m commutes with every relabeling of its carrier’s values. On two values this is commutation with the exchange. On the two-valued carrier this reading coincides with definability from equality, itself a theorem.

L1 (Arena, proved). A carrier with decidable equality admits a structureless law beyond the identity exactly when it holds two values. *[one value admits one map; three pairwise-distinct values force the identity; a moved point and its image exhaust the carrier; on two, the exchange moves both]* Marking one value leaves the relabelings that fix it, and the same equivalence holds under that weaker demand: a mark-respecting law moves the mark, or moves every value, exactly on two.

L2 (Uniqueness, proved). On two values the structureless laws are the identity and complement. The graph of the identity excludes nothing. A structureless law that excludes anything is complement.

L3 (Retorsion, proved). A tokening that marks “no distinction is drawn” differently from its denial draws one. The conjunction of a discriminated tokening of the denial with the denial is refuted.

Th. 1 (No being). $\neg\text{Fix}(\neg)$: each value differs from its own complement, and $X \leftrightarrow \neg X$ has no solution in Prop.

Th. 2 (Alternation). $\text{Run}(\neg)$; the orbit from each seed is the period-two alternation, unique given the seed, and the two orbits are relabelings of each other.

Th. 3 (Equivalence). $\neg\text{Fix}(\neg) \leftrightarrow \text{Run}(\neg)$. *[the two sides quantify over the same instances; each direction is instantiation]*

Th. 4 (Discrimination). A tokening that marks the denial of occurrence differently from its assertion witnesses occurrence, and the two tokenings are distinct. On two values, “one complement apart” and “distinct” coincide, so this is the theorem’s whole yield.

P (the premise). The tokening of the denial and the tokening of the assertion stand in oriented succession, the first succeeded by the second and the reverse failing. One instance per realized step; the full run adds an infinite supply of pairwise-distinct loci and a discrimination at every step. P is independent of Th. 1–4: two structures over one world satisfy exactly the same claims, with P false throughout one and true in the other, so no claim entails P and no claim refutes P . The independence holds relative to the world signature (T, h, tok) .

Th. 5 (Actuality, from P). Given discrimination and one oriented instance, the two loci are distinct, the successor’s value is the complement of its predecessor’s, and the structure fails to be still. A chain of oriented instances realizes the run.

Th. 6 (Obstruction). Regiment claims as properties of succession-free worlds. A claim true wherever its denial is discriminated holds at still worlds that discriminate that denial; given any such still world, the entailment from the claim to running fails. Temporal realization rests on P , and, given such a world, an undeniable substitute for P is unavailable. A still world that does discriminate is exhibited by ostension.

Th. 7 (Orientation). On loci with decidable equality, every relabeling-invariant succession is symmetric, so no invariant succession holds of a pair while failing of the reverse pair. Direction is unavailable from invariance, which is where the orientation clause of P earns its place.

Corollary. The alternation exists and is unique up to relabeling of its seed, and nothing static answers to the ground concept (Th. 1–3). A discriminated denial witnesses occurrence (Th. 4). Given P , each such denial realizes a step of the alternation (Th. 5), and P is the exact price of the temporal reading (Th. 6–7).

Ledger. (1) *Verified conditionals*: the formal results, with an empty axiom footprint. (2) *Five regimentation choices*: ground as law, structureless as equivariant, groundhood as exclusion, satisfaction as objectual or processual, claims as properties of succession-free worlds. (3) *One empirical input*: the actual world contains a tokening that discriminates the pair, discharged once by ostension; the discrimination at every step in item (4) is empirical too, and it is counted there with the premise. (4) *The premise P* : one oriented instance per realized step; the full run adds an infinite supply of pairwise-distinct loci and a discrimination at every step. (5) *A consistent constructive metalogic*.

L1–L3 and Th. 1–4 are named theorems of `lean/Becoming.lean`; Th. 5–7, P , and the independence of P and of the chain hypothesis are in `lean/Obstruction.lean`. Both files are Lean 4, prelude only, with no imports and no axioms, the standard ones included. The audits comprise 52 lines in the first file and 47 in the second.